

Ziyu (Rick) Xu

Robotics Institute, Carnegie Mellon University

Research Interests: Robotics Software, Simulation, and Reproducible Evaluation

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Education

Carnegie Mellon University (CMU)

Master of Science in Robotics (MSR)

Aug. 2026 – Expected 2028

Pittsburgh, PA

University of Michigan, Ann Arbor (UM)

B.S.E. in Mechanical Engineering (Robotics Track); Minor in Computer Science GPA: 3.85/4.0 (Top 5%)

Oct. 2024 – May 2026

- **Core Coursework:** Robot Learning, RL, Embedded Systems, Optimal Control, Convex Optimization

Shanghai Jiao Tong University (SJTU)

B.S.E. in Electrical and Computer Engineering

Sep. 2022 – Jul. 2026

GPA: 3.56/4.0 (Top 20%)

- **Core Coursework:** Algorithms, Mathematical Foundations, Probability Theory, Signals and Systems

Research Experiences and Internships

Unitree Robotics, R&D Department, Embodied AI Team

May 2026 – Present

Research Intern, Humanoid Robot Foundation Models

Hangzhou, China

- **Real-Robot Policy Infrastructure:** Built and validated a closed-loop stack on Unitree H2/G1 humanoids integrating teleoperation, VLA/WAM and whole-body policy adapters, rollout logging, evaluation, controller interfaces, and policy fine-tuning; enabled one-command launches across platforms.
- **Reproducible Evaluation:** Implemented an 8-task, 24-scene manipulation benchmark with background variation. Built PICO-based scene digitization and AR-guided reconstruction with reusable layouts, achieving 2.5 cm median placement error.

Intelligent Robotics and Autonomy Lab , UM

Oct. 2024 – Apr. 2026

Research Assistant, Directed by Prof. Vasileios Tzoumas

Ann Arbor, MI

- **Multi-Robot Simulation and ROS2:** Rebuilt a Unity + AirSim multi-drone testbed with revised dynamics and sensing; migrated SlideSLAM communication from ROS1 to ROS2 for bandwidth-limited active-SLAM evaluation.

Computational Autonomy and Robotics Lab, UM

Jun. 2025 – Apr. 2026

Research Assistant, Directed by Prof. Maani Ghaffari

Ann Arbor, MI

- **Sensor Integration:** Designed mounts and integrated, aligned, and calibrated GPS, lidar, IMU, and cameras for field data collection and closed-loop control.
- **Learning-Augmented Lie-MPC:** Developed and field-tested an autonomous surface vehicle controller that estimates residual wind/current disturbances online and corrects the dynamics used by MPC. Achieved a 20% improvement in trajectory-tracking performance over baseline MPC in real-ASV experiments.

Ziyu (Rick) Xu

Intelligent Control Lab, Carnegie Mellon University

Graduate Student Researcher, advised by Prof. Changliu Liu

Research experience continued

Aug. 2026 – Present

Pittsburgh, PA

- **Whole-Body and Safe Locomotion Research (ongoing):** Investigating whole-body control, safety with control barrier functions (CBFs), and world-model-based locomotion.
- **Dexmate Bimanual Manipulation Demo, IROS 2026:** Contributed to a bimanual block-stacking demonstration combining agent planning with policy execution. Implemented the agent component, prepared the on-site environment, adapted the system to different grippers, and performed system calibration.

Shanghai Artificial Intelligence Lab & IRMV, SJTU

Research Assistant, Directed by Prof. Junchi Yan and Prof. Hesheng Wang

Apr. 2026 – Present

Shanghai, China

- **Iterative Visual KV Compression:** Studied errors from reusing retained visual KV entries after neighboring tokens are removed in multi-turn and streaming inference. Developed context-aware re-encoding at original positions and compared with inheritance and restoration under matched memory and runtime budgets.

Publications and Manuscripts

- Yinan Dong, **Ziyu Xu**, Tsimafei Lazouski, Sangli Teng, and Maani Ghaffari. “Online Learning-Enhanced Lie Algebraic MPC for Robust Autonomous Surface Vehicle Control”, *IEEE Robotics and Automation Letters (RA-L)*, revised manuscript under review, 2025.
- Xiaohan Wang^{*}, **Ziyu Xu**^{*}, Xuyi Yang. “Re-encoding What You Keep After Every Compression: Retained Visual KV Is Computed in a Context That No Longer Exists”, *Submitted to ICLR 2027*. ^{*}*Equal contribution*.
- Xuyi Yang, Xiaohan Wang, Wenhao Zhang, and **Ziyu Xu**. “StreamJIT: Continuous Observation and Just-in-Time Deliberation for Proactive Streaming Video Assistants”, *Submitted to ICLR 2027*.
- C. Yuan, J. Jiang, K. Yang, **Z. Xu**, . . . , and Y. Ban. “Systematic Evaluation and Guidelines for Segment Anything Model in Surgical Video Analysis”, *npj Digital Surgery*, accepted, 2024.

Selected Awards

Gold Medal, University Physics Competition (2023); Dean’s Honor List, University of Michigan (2024, 2025).

Skills

- **Robotics Systems:** ROS/ROS2, Python/C++, Unity + AirSim, Linux, Docker, Git, sensor integration, teleoperation and policy interfaces
- **Evaluation and ML:** Real-robot benchmarks, rollout logging, PyTorch, OpenCV, scene reconstruction, model evaluation under matched budgets